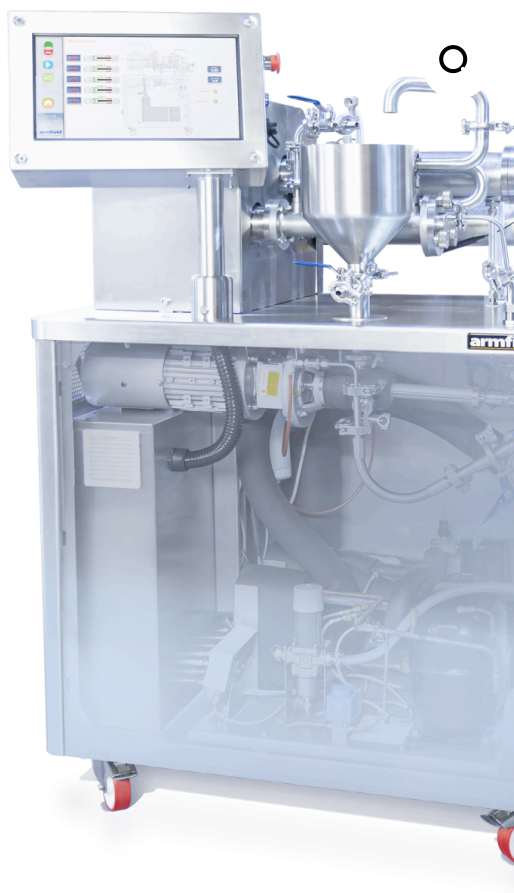




FT25 Scraped Surface Heat Exchanger Series

The FT25 range of miniature-scale scraped surface heat exchangers has built-in control and flexibility with a graphic representation of the process on the touch screen control panel. The operator can choose the parameters required for any particular formulation, these include **margarine**, **shortening** and **ice cream** processes.

The FT25 has a number of configurations according to the required use.

Armfield miniature-scale technology is well established at offering developers the opportunity to run small trials, which provide enough information to enable scale-up to larger plant with confidence.

The formulations go through similar physico-chemical changes as they would in production plant and real time data logging permits results to be recorded and repeated.

PRODUCT CODE: FT25

Categories: Armfield Edible And Essential Oils, Armfield Educational Food Technology, FT Series - Educational Food Technology, FT25X Modular Scraped Surface Heat Exchanger Systems, IFT Industrial Food Technology, IFT Series - Industrial Food Technology

Tags: 20 Bar , 40 Bar , Barrel , Crystalliser , FT25 , FT25XA , FT25XA Scraped Surface Heat Exchanger , Heat Exchanger , Ice Cream , Margarine , Pinworker , Refrigeration , Scraped , Scraped Surface

DESCRIPTION

The FT25 is a lab-scale crystalliser with scraped surface heat exchangers mounted on a stainless-steel frame supplied with a large 15.4 inch HD touchscreen and PLC control system. The working platform can accommodate one or two barrels (B)

and a variable speed pinworker (P) alongside the hopper and connection ports.

The unit is housed in a Stainless Steel (304) cabinet with removable sides for easy access. It uses a refrigeration system, supplied within the cabinet with a chilled, food-grade refrigerant, which is pumped through the barrel jackets. An air incorporation system with pump and control can also be added if required. Enhanced CIP system is also available. Small batch sizes enable rapid product testing saving time, ingredients and utility costs.

Small batch sizes enable rapid product testing. Enhanced CIP facility available (FT52).

Note: For Scraped surface heat exchanger applications involving product heating, UHT, or combined heating and cooling, please enquire about our FT174XA – Modular Miniature HTST/UHT Process System.

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TECHNICAL SPECIFICATIONS

HEAT EXCHANGER BARREL

Heat transfer area: 0.045m²

Shaft/barrel volume ratio: 1 : 3.5

Scraper blades: 4 PEEK scrapes along stainless steel shaft, 2 pairs with 180° separation

Barrel length: 0.35m

Barrel working pressure: 20 bar or 40 bar

PINWORKER

No of pins on shaft: 36

No of pins on barrel: 36

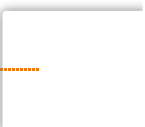
BARREL/PINWORKER DRIVES

Variable speed ac motors, inverter controlled

Speed range: 100-1400rpm

REFRIGERATION SYSTEM

Food grade refrigerant: -25°C capability (Coolflow LVF50)



FEED PUMPS

Progressive cavity pump

Process throughput: Up to 0-20l/h, independent of back pressure

Particulate capability: up to 0.6mm

Fibre handling capability: up to 25mm

Air incorporation system: Air Pump with fine filter and digital flow controller

NOTE: Air supply is injected into the inlet of the feed pump.

FEATURES & BENEFITS

- Duplicates full-scale process
- Fast, accurate new product development
- All process parameters under operator control for maximum flexibility
- Rugged and reliable units
- Only requires electricity and water to function
- Full control of barrel speeds to over 1400rpm
- Temperature control system incorporated
- Integral control panel enabling simple control and monitoring of all major variables
- Enclosed self-contained, mobile, stainless steel service cabinet for easy cleaning and wash down with removable panels
- Stainless steel barrel with PEEK scrapers
- Hygienic design
- Made from 316 stainless steel
- In cabinet refrigeration

Available:

- Variable speed pinworker for margarine and soft spreads
- Air incorporation system
- Computer data logging facility
- Integrated **CIP** Pump

CONTROL PANEL

The FT25 utilises a cleanable touchscreen HMI with intuitive displays.

All process parameters can be accessed and changed via the process HMI screen including coolant temperature, individual motor On/Off

and rotational speeds and, where fitted, air incorporation flowrate.

The process screen also incorporates key alarms to inform the operator of events preventing operation such as over pressure.

Realtime data is displayed from sensors and other devices providing the operator with instantaneous temperatures, pressure as well as motor currents and RPMs.

A graphing screen can also be called up where individual outputs can be displayed on a graph.

Note: For Scraped surface heat exchanger applications involving product heating, UHT, or combined heating and cooling, please enquire about our FT174 – Modular Miniature HTST/UHT Process System.

Software



DATA CAPTURE SOFTWARE

The FT25 can data log all operational parameters when activated by the operator. The data will be stored in discrete data files which can be downloaded to a USB stick to allow for offline analysis using spreadsheets.

Supplied as standard, the FT25 can log data.

Typical Configurations

Terminology: Throughout this document, barrel refers to a **scraped surface heat exchanger (SSHE)**.

CONTINUOUS MARGARINE CRYSTALLISER

FT25-BP-20: One barrel and one pinworker.

FT25-BBP-20: Two barrels and one pinworker. A resting tube for product settling and food grade refrigeration unit for process cooling are supplied.

A range of oil emulsions can be processed.

CONTINUOUS ICE CREAM FREEZER

FT25-BA-20: One barrel Air incorporation via a small air compressor and refrigeration. Overrun can be achieved up to 300% (dependant on product).

COMBINED ICE CREAM AND MARGARINE SYSTEM

FT25-BBPA-20: Two barrels, one pinworker plus air incorporation and refrigeration.

Available Configurations

In addition to being available in 20bar and 40bar setups the following process configurations are available:

- Barrel with Air Incorporation
- Barrel and Pinworker
- Two Barrels and Pinworker
- Two Barrels and Pinworker with Air Incorporation

Terminology: Throughout this document, barrel refers to a **scraped surface heat exchanger (SSHE)**.

Control Console – HMI

The FT25 utilises a cleanable touchscreen HMI with intuitive displays. All process parameters can be accessed and changed via the process HMI screen including coolant temperature, individual motor On/Off and rotational speeds and, where fitted, air incorporation flowrate. The process screen also incorporates key alarms to inform the operator of events preventing operation such as over pressure.

Realtime data is displayed from sensors and other devices providing the operator with instantaneous temperatures, pressure as well as motor currents and RPMs.

A graphing screen can also be called up where individual outputs can be displayed on a graph.

Requirements

Electricity supply: (see ordering codes)

Water supply: For refrigeration cooling Cooling water supply with flow rate of 3l/min @ 3bar and $\leq 20^{\circ}\text{C}$



Shipping Specifications

Packed and created shipping specifications

Volume: 2.1m³

Mass: 390Kg

Dimensions

Length: 1.01m

Width: 0.72m

Height: 1.45m

Order Codes

- **FT25-20BAR-A** 220-240v 1ph 50Hz (30 A max)
- **FT25-20BAR-G** 220-240v 1ph 60Hz (30 A max)
- **FT25-40BAR-A** 220-240v 1ph 50Hz (30 A max)
- **FT25-40BAR-G** 220-240v 1ph 60Hz (30 A max)

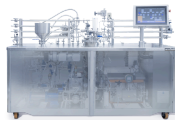
Some configurations are available with a maximum pressure rating of up to 40 bar. Please contact your Armfield representative for further details.

YOU MAY ALSO LIKE...



**FT74XA
HTST/UHT
SYSTEM**

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**FT174 –
MODULAR
MINIATURE
SCALE
HTST/UHT
PROCESS
SYSTEM**

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**FT18-MKII
MODULAR
CROSS FLOW
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**FT140X –
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VESSELS**

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